Cardiovascular System (CARD) · NPLEX-P1-CARD-16

CV-16 Immune Responses in Cardiovascular Disease

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-16-01-C01				
NPLEX-P1-CARD-16-01-C02				
NPLEX-P1-CARD-16-01-C03				
NPLEX-P1-CARD-16-01-C04				
NPLEX-P1-CARD-16-01-C05				
NPLEX-P1-CARD-16-01-C06				
NPLEX-P1-CARD-16-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C01-BD1

Rebuild atheroma formation as an inflammatory response

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-BD1 Date: _____

Without looking at your notes, draw the sequence that turns a healthy artery into a fibrofatty plaque, from endothelial dysfunction through to a cap over a necrotic core. Beside each step write the molecules doing the work: the transcription factor the injured endothelium activates, the two adhesion molecules and the chemokine it induces, the cytokine that turns monocytes into macrophages, and the growth factors that build the cap. Write which white cells arrive first and why the adhesion molecule involved selects them. Then write what happens to LDL once it is retained in the intima, which receptors carry it into the macrophage, why those receptors do not switch off as cell cholesterol rises, which enzyme stores the cholesterol, and which transporters move it back out. Finish by naming the earliest visible lesion and the age group it already appears in, and list the sites in the arterial tree where plaques form first with the flow reason.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C01-BD2

Why the smaller plaque is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-BD2 Date: _____

Two coronary lesions sit in the same artery. Lesion A narrows the lumen by about 40%, has a large lipid core, a thin cap, and dense macrophages at its edges. Lesion B narrows it by about 85% under a thick collagenous cap with many smooth muscle cells and few inflammatory cells. For each lesion write what the patient is likely to experience and which one is more likely to cause an infarction, then explain the difference from the enzymes and cytokines acting on cap collagen. Name the enzymes that degrade the collagen, the cytokine that stops smooth muscle cells replacing it, and what happens to those smooth muscle cells. Write where on the plaque rupture usually begins and why, what is exposed when it tears, and how quickly thrombus follows. Then write the other surface event that can cause coronary thrombosis without a tear in a cap. Finish by explaining why a patient with rheumatoid arthritis or lupus develops coronary disease faster, naming the cytokines that connect the two, and state what a high sensitivity CRP of 4 mg/L tells you about this patient and the cytokine route that produced that number.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-16-01-C01-RT01

Rebuild the atheroma sequence and the inflammasome branch

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C01 Activity: NPLEX-P1-CARD-16-01-C01-RT01 Date: _____

Rebuild the eight-step sequence of atheroma formation from memory, one step per line and in order, from the first insult to the formation of thrombus. Beside the step where monocytes enter, write the two molecules that made them stick and the one that drew them in. Beside the foam cell step, write the receptor family involved and why uptake has no brake. Then, underneath the sequence, draw a second short branch coming off the foam cell step: start with oxidized LDL on Toll-like receptors, add what the macrophage makes ready, then what cholesterol crystals do inside the cell, which complex assembles, which enzyme it activates, which cytokine is released in its active form, which second cytokine follows, and what the liver then produces. Write the three risk bands for that liver protein with their values.

Rebuild from memory: Atheroma formation as a chronic inflammatory response

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C02-BD1

Rheumatic fever from the throat to the valve

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C02 Activity: NPLEX-P1-CARD-16-01-C02-BD1 Date: _____

Write out acute rheumatic fever from memory as a sequence of six steps, beginning with the infection and ending with the chronic valve lesion. Name the organism, the site of infection that leads to this disease, the usual age range, and the length of the gap between infection and carditis. Name the two bacterial antigens involved and, for each, the host protein it resembles or binds, then write in two sentences why antibodies and T cells raised against the bacterium attack the heart. Write which hypersensitivity type the disease is assigned to and which cell type actually drives the long-term valve damage. Then describe the diagnostic microscopic lesion: name it, name the altered macrophage inside it, and write which layers of the heart wall contain it and what that means for the extent of the carditis. Finish by writing what the acute valve lesions look like, their size, where on the leaflet they sit, which valve is affected most, and what repeated attacks turn the mitral valve into.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C02-BD2

Work a suspected case from timing and serology

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C02 Activity: NPLEX-P1-CARD-16-01-C02-BD2 Date: _____

An 11-year-old girl has fever of 38.7 degrees C (101.7 degrees F), migratory pain and swelling of both knees and one elbow, a new murmur, an ESR of 72 mm/h, and a PR interval of 0.24 s. Her throat culture is negative. Using the Jones criteria, write which of her findings are major and which are minor, state whether the criteria are met for an initial attack and show the counting, and write which of her findings cannot be counted here and why. Write how you would prove the recent streptococcal infection now that the culture is negative, naming both antibody tests, when each rises and peaks, which one stays up longer, and why a rising value is better evidence than one measurement. Then write why her joints will recover completely while her valve may not. Finally, a different child presents months later with involuntary writhing movements and no fever, and another child develops skin sores rather than a sore throat and later has dark urine and red cell casts. For each child write which disease fits, the immune mechanism behind it, and why the site of the original infection decides the answer.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C03-BD1

Organize the vasculitides by immune effector

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C03 Activity: NPLEX-P1-CARD-16-01-C03-BD1 Date: _____

On blank paper sort the vasculitides from this chapter into three groups by the immune effector that injures the vessel: immune complex disease, ANCA disease, and granulomatous T-cell disease. List the diseases under each heading. For the immune complex group write how the complexes form, where they deposit and why those sites, which complement pathway they use, which complement fragment brings in neutrophils, what the biopsy shows on immunofluorescence, what the dying neutrophils leave behind and what that appearance is called, and what the lesion looks like in the skin. For the ANCA group write the two target enzymes with their staining patterns, why one of them stains the nucleus, what primes the neutrophil, the two ways the antibody engages the same cell, what the neutrophil then releases, and why the biopsy shows almost no deposits. For the granulomatous group write which cells start the reaction, which cytokine recruits the macrophages, which layer of the artery they gather in, what happens to the internal elastic lamina and the intima, and which cytokine drives the very high ESR.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C03-BD2

Name the vasculitis from complement and serology

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C03 Activity: NPLEX-P1-CARD-16-01-C03-BD2 Date: _____

Four patients have palpable purpura. Patient 1 is 52, has a C4 that is very low with a near normal C3, rheumatoid factor, and hepatitis C antibody. Patient 2 is 61, has normal complement, hemoptysis, red cell casts, and MPO-ANCA. Patient 3 is 8 years old, had a cold two weeks ago, has abdominal pain, arthralgia, normal complement, and mesangial IgA. Patient 4 is 47, has hypertension, abdominal pain after eating, a foot drop with a wrist drop on the other side, hepatitis B surface antigen, and negative ANCA with clear lungs. For each patient name the disease, write the immune mechanism, and write the one laboratory or biopsy finding that settles it. Then explain in one or two lines each why C4 is very low in patient 1, why complement is normal in patient 3 even though complexes are deposited, why patient 2 has no deposits on immunofluorescence, and why patient 4 has renal infarcts and renin-dependent hypertension but no glomerulonephritis. Finish with two more patients: a 74-year-old with a temporal headache, pain on chewing, sudden vision loss in one eye, and an ESR of 105 mm/h, and a 29-year-old woman with a weak left arm pulse and a 30 mm Hg difference in arm blood pressures. Write what each has, the shared mechanism, the vessels involved in each, and which artery occlusion causes the vision loss.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-16-01-C03-CP01

Immune complex, ANCA, and granulomatous vasculitis side by side

Fill in every cell of the table from memory, then check it against the notes and correct what you missed.

Feature	Immune complex vasculitis	ANCA vasculitis	Granulomatous large vessel arteritis
Hypersensitivity type and immune effector cell or molecule			
What immunofluorescence of the biopsy shows			
What serum complement does and why			
Serum marker or antibody to order			
Vessel sizes and tissues usually involved			
Example diseases from this chapter			
Typical skin, kidney, and lung findings			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C04-BD1

Three immune routes into the heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-BD1 Date: _____

Write the three conditions in this section as three short blocks on one page: delayed pericarditis after infarction, the sterile valve lesion of lupus, and heart block in a newborn. For each one write how the antigen or antibody reaches the heart, the immune mechanism, and the tissue injured. For the pericarditis, write how long after the infarct it appears and how that differs in timing and mechanism from the pericarditis of the first few days, then list the symptoms, the examination finding, the blood results, and what the ECG shows and how that pattern differs from the ECG of the infarct itself. Name the operation that produces the same syndrome. For the valve lesion, write what the vegetations are made of, which antibody promotes the thrombosis, which surfaces of the leaflet they sit on, which valve is involved most often, what they leave behind as they heal, and where fragments travel. For the newborn, name the maternal antibody and the second one often found with it, the receptor that carries it across the placenta, the trimester it starts, the maternal diseases to ask about, and the structure destroyed.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C04-BD2

Explain the newborn and sort four sets of vegetations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-BD2 Date: _____

A newborn has a ventricular rate of 50 per minute, P waves at 142 per minute that bear no relation to the QRS complexes, and annular pink lesions on the face. The mother has no symptoms. Write what the atria and ventricles are each doing and what the ventricular rhythm is called, then explain the two separate ways the maternal antibody slows conduction, one immediate and one permanent, and name the cytokines released by the macrophages. Write the window in gestation when the node is injured and why the ion channel the antibody blocks matters at this particular site. Then write what has happened by 8 months to the rash and to the bradycardia, and explain each outcome. Write the risk of this complication in an antibody positive pregnancy and the risk after one affected child. Finish by taking four patients with vegetations on the mitral valve: one had streptococcal pharyngitis as a child, one has lupus with antiphospholipid antibodies, one has fever with positive blood cultures, and one has advanced pancreatic cancer. For each write the cause, the size and position of the vegetations on the leaflet, whether organisms are present, and what the valve looks like years later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-16-01-C04-RT01

Rebuild the pathway to congenital heart block

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C04 Activity: NPLEX-P1-CARD-16-01-C04-RT01 Date: _____

Rebuild the six-step pathway of congenital heart block from memory, one step per line and in order, from the maternal antibody to the permanent rhythm. Beside the placental step name the receptor that carries the antibody and the trimester transfer begins. Beside the binding step write what the antibody binds on the fetal myocyte and which ion channels it also inhibits. Beside the macrophage step name the two mediators released. Beside the final step write the ventricular rate you would expect against the normal fetal rate range. Then write one line under the sequence saying which step makes the block permanent and why that tissue cannot recover.

Rebuild from memory: Pathogenesis of congenital heart block

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C05-BD1

Build the rejection table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C05 Activity: NPLEX-P1-CARD-16-01-C05-BD1 Date: _____

On blank paper build a table of the four types of heart transplant rejection with one row each, using the columns timing after transplant, immune effector, hypersensitivity type, and findings in the graft, and fill every cell. Above the table write the two ways recipient T cells see donor HLA: for each, write whose antigen presenting cells are involved, where the HLA peptides sit, which one drives the strong early response and why, and which one keeps working after the donor cells carried in the graft are gone and gives B cells the help to make antibody. Beneath the table write, for each type, the one feature that identifies it: what the preformed antibody does to small vessels, what the biopsy shows when T cells are the effector, which complement fragment marks the capillaries in antibody-mediated rejection and why it stays bound, and what the coronary arteries look like in the chronic form. Finish with the two reasons hyperacute rejection is now rare.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C05-BD2

Three recipients and a denervated heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C05 Activity: NPLEX-P1-CARD-16-01-C05-BD2 Date: _____

Three heart transplant recipients come for review. Recipient 1, at 3 weeks, feels well; biopsy shows lymphocytes around vessels and between damaged myocytes with no C4d. Recipient 2, at 5 months, is hypotensive and breathless; biopsy shows swollen capillary endothelium with macrophages inside capillaries, strong C4d, few interstitial lymphocytes, and a new donor-specific anti-HLA antibody has appeared. Recipient 3, at 6 years, has breathlessness with no chest pain, and every coronary branch is smoothly and evenly narrowed. For each recipient name the type of rejection, write the effector, and explain why the timing fits that effector. Write why recipient 2 is more unwell than recipient 1 despite a similar biopsy grade. For recipient 3, write how the artery wall differs from ordinary atherosclerosis in distribution, in the elastic lamina, and in lipid and calcium content, name the immune and the nonimmune drivers, and write why angiography understates the narrowing. Then write the resting heart rate you expect in all three and why, how the rate responds to exercise and through what route, and why recipient 3 has no angina and what he presents with instead. Finish by writing what a graft that thromboses within minutes of reperfusion would mean and which antibodies cause it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C06-BD1

Viral myocarditis in three phases

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C06 Activity: NPLEX-P1-CARD-16-01-C06-BD1 Date: _____

Write the definition of myocarditis in one sentence. Then rebuild viral myocarditis from memory as three numbered phases. For phase 1 name the classic virus, the receptor and co-receptor it uses to enter the myocyte, the viral enzyme that cuts a cytoskeletal protein and which protein that is, the time frame, and the innate defences that limit spread. For phase 2 write the time frame, which T cells kill infected myocytes and with which molecules, which cytokines the helper cells release, why this phase causes most of the muscle loss even though it clears the virus, and which released cardiac protein can set off autoimmunity and by what mechanism. For phase 3 write what happens to most patients and what a minority develop, with the two reasons it happens. Then write how the diagnosis is made: name the histological criteria and their weakness, and write where the MRI signal sits in the wall and how that differs from infarction. Finish with the clinical findings and explain each from the pathology: the blood marker, the chest pain with ST changes, the heart failure, and the arrhythmias that can kill a young athlete.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C06-BD2

Separate four patients with myocarditis

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C06 Activity: NPLEX-P1-CARD-16-01-C06-BD2 Date: _____

Four patients have myocardial inflammation. Patient 1 is 22 and had fever and sore throat ten days ago, now with chest pain, a troponin of 4.2 ng/mL, and ST changes crossing more than one coronary territory. Patient 2 started a sulfonamide two weeks ago and has a rash, fever, a blood eosinophil count of 2.4 x 10 to the 9 per L, and a falling ejection fraction. Patient 3 has long standing asthma, an eosinophil count of 9 x 10 to the 9 per L, a numb foot, and a thickened endocardium with mural thrombus. Patient 4 is 17, male, and developed chest pain three days after a second dose of an mRNA vaccine, with a mildly raised troponin and subepicardial enhancement on MRI. For each patient write the form of myocarditis, the effector cell or molecule doing the damage, and what you expect the biopsy to show. Write which patient is most likely to end with restrictive physiology and explain the route from eosinophil granule to that physiology. Write what you expect to happen to patient 2 once the drug is stopped and why, and state whether the reaction depends on the dose. For patient 4 write what the notes say about the mechanism and the usual course. Then add a fifth patient whose heart failure worsens over two weeks with wide necrosis and multinucleated cells on biopsy and no granulomas: name the form, its associations, and its outlook.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-16-01-C06-PR01

Predict what the T-cell phase does to the myocardium

A 24-year-old man is in the second week after a coxsackievirus B infection of the myocardium, and the CD8 T-cell response is now at its peak. Predict each item below, write the direction of the change, and give the cell, molecule, or structural event that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the amount of virus in the myocardium		
What happens to the number of surviving myocytes compared with the first few days		
What happens to the serum troponin		
What happens to the integrity of dystrophin and the myocyte membrane		
What happens to circulating exposure to cardiac myosin and to the chance of autoimmunity		
What happens to ventricular chamber size and wall thickness over the following months in the minority who do not recover		
What happens to the risk of reentrant ventricular arrhythmia as fibrosis replaces lost muscle		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-16-01-C07-BD1

HHV-8 latency and the cytokines of myxoma

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C07 Activity: NPLEX-P1-CARD-16-01-C07-BD1 Date: _____

Divide a page in two. On the left, write Kaposi sarcoma from memory: name the virus and its family, how it spreads, which cell it infects latently and what type of endothelium that cell resembles. Then list the viral genes and what each one does, covering the protein that holds the viral genome in the cell and the two tumour suppressors it blocks, the viral cyclin and the partner kinase it drives, the protein that blocks apoptosis and the transcription factor it turns on, the receptor that induces VEGF, and the viral cytokine and the receptor chain it does not need. Write what the lesions look like under the microscope, the three stages they pass through, the tissues involved, and the stain that confirms the virus. Name two other diseases the same virus causes. On the right, write atrial myxoma: its rank among primary cardiac tumours in adults and in children, which chamber and which spot in it, what the mass looks like and what cells and matrix it contains, the usual patient, and the inherited syndrome with its gene and two accompanying features.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-16-01-C07-BD2

Failed surveillance, and one cytokine imitating three diseases

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-16 Immune Responses in Cardiovascular Disease
Competency: NPLEX-P1-CARD-16-01-C07 Activity: NPLEX-P1-CARD-16-01-C07-BD2 Date: _____

Write which cells normally hold the latent virus of Kaposi sarcoma in check and which helper cells they depend on. Then take three patients: one with untreated HIV and a CD4 count of 90 cells per microlitre, one taking a calcineurin inhibitor after a kidney transplant, and an 68-year-old man of Mediterranean descent with indolent plaques on both shins. For each write why the tumour appeared, how aggressive it is, and what you expect to happen to the lesions if immunity is restored or immunosuppression is reduced, including the transient change that can occur first and which lesions may not respond. Write which viral protein in HIV adds to the proliferation. Then move to a 54-year-old woman with three months of fever, weight loss, fatigue, an ESR of 88 mm/h, a haemoglobin of 9.8 g/dL, raised immunoglobulins across all classes, and breathlessness and dizziness that come on when she stands up, with a sound in early diastole. Write which single cytokine explains the systemic picture and trace it to each finding: the fever, the high ESR, the anaemia, and the raised immunoglobulins, naming the mediator or protein at each step. Explain the positional symptoms and the diastolic sound from where the mass sits, write what else fragments of it can do, and write what happens to all of the systemic findings after the mass is removed and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-16-01-C07-CA01

Fever, weight loss, and a high ESR in a woman with positional breathlessness

A 56-year-old woman is referred after four months of intermittent fever, 7 kg of weight loss, and aching joints. She also describes breathlessness and lightheadedness that come on within seconds of standing and settle when she lies down, and she has had two brief episodes of weakness of the right arm. Temperature is 38.2 degrees C (100.8 degrees F), pulse is 96/min and regular, and blood pressure is 118/72 mm Hg. On auscultation there is a low-pitched sound shortly after the second heart sound, and the intensity of a diastolic murmur changes when she sits up. Haemoglobin is 9.7 g/dL with a mean corpuscular volume of 84 fL, ferritin is 260 ng/mL, ESR is 92 mm/h, and serum immunoglobulins are raised in all classes. Three sets of blood cultures are sterile, antinuclear antibody and ANCA are negative, and complement is normal. Transthoracic echocardiography shows a mobile pedunculated mass attached to the interatrial septum near the fossa ovalis that moves toward the mitral orifice during diastole.

1. Recognize. Name the lesion, the chamber it arises in, and the attachment site that the echocardiographic description identifies.

2. Explain. Name the cytokine the lesion secretes and trace it to the fever, the high ESR, the anaemia, and the raised immunoglobulins, naming the mediator or liver protein at each step.

3. Differentiate. Write how this picture differs from infective endocarditis, from a systemic vasculitis, and from lupus, naming the findings in this patient that separate them.

4. Investigate. Write which features of the mass and the blood work you would examine to support the diagnosis, and what the histology and the inheritance history would show.

5. Decide. State what is causing the positional breathlessness and the episodes of arm weakness, and name the microscopic feature of the mass that permits the second of these.

6. Justify. Justify your answer by explaining why a benign tumour produces a picture resembling chronic infection or malignancy, and what you expect to happen to the fever, anaemia, ESR, and immunoglobulins after the mass is removed.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-16-01-C02-Q01

Carditis three weeks after a sore throat

A 10-year-old boy is brought to the physician because of fever and painful swelling that began in the right knee and moved to the left ankle. He had a sore throat 3 weeks ago that was not treated. Temperature is 38.6 degrees C (101.5 degrees F). A new apical systolic murmur is heard. ESR is 78 mm/h, throat culture is negative, and the antistreptolysin O titer is rising. Which of the following best explains the cardiac findings in this patient?

A. Bacterial invasion of the valve leaflets with destruction of the endocardial surface

B. Antibodies and T cells raised against streptococcal M protein and group A carbohydrate that also recognize cardiac myosin and valve proteins

C. Deposition of circulating streptococcal antigen and antibody complexes in the valve with classical pathway activation

D. Antineutrophil cytoplasmic antibodies activating primed neutrophils on the endocardial surface

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C03-Q02

Purpura, casts, and a very low C4

A 58-year-old woman comes to the physician because of raised purple lesions on both lower legs that do not blanch, aching wrists, and numbness of both feet. Serum C4 is markedly decreased with a near normal C3, rheumatoid factor is present, and hepatitis C antibody is positive. Urinalysis shows red cell casts. Skin biopsy shows granular immunoglobulin and complement in the walls of dermal venules. Which of the following is the mechanism of vascular injury in this patient?

A. IgM rheumatoid factor bound to IgG forming complexes that deposit in vessel walls and consume the classical pathway

B. Antibodies against myeloperoxidase activating primed neutrophils with little immunoglobulin deposition

C. Antibody binding to type IV collagen in the vessel basement membrane

D. Dendritic cell activation of CD4 T cells with macrophage granuloma formation in the vessel media

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C05-Q03

Hypotension five months after transplantation

A 49-year-old man who received a heart transplant 5 months ago is admitted because of 2 days of breathlessness. Blood pressure is 84/56 mm Hg. Endomyocardial biopsy shows swollen capillary endothelial cells with macrophages within capillary lumens, strong capillary staining for C4d, and only scattered interstitial lymphocytes. A donor-specific anti-HLA antibody that was absent before transplantation is now detected. Which of the following best accounts for these findings?

A. CD8 T cells recognizing intact donor HLA on passenger dendritic cells

B. Smooth muscle migration into the intima driven by PDGF and TGF-beta

C. Donor-specific antibody binding capillary endothelium with classical pathway activation

D. Preformed anti-ABO IgM causing immediate small vessel thrombosis

Why did you pick it?

Board-style question NPLEX-P1-CARD-16-01-C06-Q04

A young man two weeks after a viral illness

A 25-year-old man comes to the emergency department because of 2 days of chest pain and breathlessness. He had fever, myalgia, and diarrhoea 12 days ago. Troponin I is 5.6 ng/mL, and the electrocardiogram shows ST segment changes in leads that span more than one coronary territory. Coronary angiography is normal. Cardiac MRI shows late gadolinium enhancement in the subepicardial layer of the lateral wall. Which of the following is most responsible for the myocyte loss at this point in his illness?

A. Cleavage of dystrophin by viral protease 2A

B. Eosinophil major basic protein released into the endocardium

C. Subendocardial necrosis from occlusion of an epicardial coronary artery

D. CD8 T cells killing infected myocytes with perforin and granzymes

Why did you pick it?